## **Lecture 28** (*April 28*)

*Reading: Section 7.4*
*Viewing: Lecture 33*

In practice, the singular value decomposition (241) of a matrix gives us a computationally manageable way to compute how $A$ acts on arbitrary vectors:

$$\boldsymbol{v} = c_1\boldsymbol{v}_1 + \cdots + c_n\boldsymbol{v}_n \tag{250}$$

(since the $\boldsymbol{v}_i$ are orthonormal, we have $c_i = \boldsymbol{v}_i^T\boldsymbol{v}$). Then we have:

$$A\boldsymbol{v} = \sigma_1 c_1 \boldsymbol{u}_1 + \cdots + \sigma_r c_r \boldsymbol{u}_r \tag{251}$$

It is customary to order your singular values $\sigma_1 \geq \cdots \geq \sigma_r > 0$. In this case, you can compare (250) with (251) to get:

$$\frac{||A\boldsymbol{v}||}{||\boldsymbol{v}||} = \sqrt{\frac{\sigma_1^2 c_1^2 + \cdots + \sigma_r^2 c_r^2}{c_1^2 + \cdots + c_r^2 + \cdots + c_n^2}} \leq \sigma_1$$

Moreover, we get equality in the inequality above when $\boldsymbol{v}$ is a multiple of $\boldsymbol{v}_1$. Therefore:

$$\sigma_1 \textbf{ is the maximum of } \frac{||A\boldsymbol{v}||}{||\boldsymbol{v}||}, \textbf{ achieved for } \boldsymbol{v} = \boldsymbol{v}_1 \tag{252}$$

which is an intrinsic characterization of the largest singular value. Specifically, we are saying that the largest singular value is equal to the **norm** of the matrix:

$$||A|| := \max_{\boldsymbol{w}\in\mathbb{R}^n\setminus 0} \frac{||A\boldsymbol{w}||}{||\boldsymbol{w}||} \in \mathbb{R}_+ \tag{253}$$

The norm of a matrix satisfies the usual "triangle inequality" satisfied by the norms of vectors:

$$||A + B|| \leq ||A|| + ||B|| \tag{254}$$

(proof: for any vector $\boldsymbol{w}$, we have $||(A+B)\boldsymbol{w}|| \leq ||A\boldsymbol{w}|| + ||B\boldsymbol{w}|| \leq ||A|| \cdot ||\boldsymbol{w}|| + ||B|| \cdot ||\boldsymbol{w}|| = (||A|| + ||B||) \cdot ||\boldsymbol{w}||$, which implies (254)) as well as a similar inequality when multiplying matrices:

$$||AB|| \leq ||A|| \cdot ||B|| \tag{255}$$

(proof: for any vector $\boldsymbol{w}$, we have $||AB\boldsymbol{w}|| \leq ||A|| \cdot ||B\boldsymbol{w}|| \leq ||A|| \cdot ||B|| \cdot ||\boldsymbol{w}||$, which implies (255)). You can obtain norm-like descriptions of the other singular values of a matrix, as follows:

$$\sigma_2 \textbf{ is the maximum of } \frac{||A\boldsymbol{v}||}{||\boldsymbol{v}||} \textbf{ among those } \boldsymbol{v} \perp \boldsymbol{v}_1, \textbf{ achieved for } \boldsymbol{v} = \boldsymbol{v}_2 \tag{256}$$

$$\sigma_3 \textbf{ is the maximum of } \frac{||A\boldsymbol{v}||}{||\boldsymbol{v}||} \textbf{ among those } \boldsymbol{v} \perp \boldsymbol{v}_1, \boldsymbol{v}_2, \textbf{ achieved for } \boldsymbol{v} = \boldsymbol{v}_3 \tag{257}$$

etc. In fact, this recursive procedure <u>defines</u> the singular values $\sigma_i$ and the right singular vectors $\boldsymbol{v}_i$ (from which the left singular vectors can be deduced from (243)): first $\sigma_1$ and $\boldsymbol{v}_1$ are defined by (252), then $\sigma_2$ and $\boldsymbol{v}_2$ are defined by (256), then $\sigma_3$ and $\boldsymbol{v}_3$ are defined by (257) etc.

Geometrically, the SVD in the form (244) means that any matrix $A$ can be factored as:

$$\Big(\text{rotation}\Big)\Big(\text{scaling}\Big)\Big(\text{another rotation}\Big) \tag{258}$$

Compare this with (233) for symmetric matrices: the novelty in (258) is that the two rotations, which correspond to the orthogonal matrices $U$ and $V^T$ respectively, are now completely different linear transformations acting on two different vector spaces. This is natural, since $A$ corresponds to a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$. Meanwhile, the two rotations in (233) are inverses to each other, which is natural because the symmetric matrix therein had the same domain and codomain.

You can find a picture of the factorization (258) on page 392 of the textbook: the two rotations do not change any given shape, they just rotate it around space. But the scaling in the middle does change shapes by dilations across various axes: circles become ellipses, squares become rectangles etc. In the case of $2 \times 2$ matrices, the factorization (258) explicitly says that:

$$\underbrace{\begin{bmatrix} a & b \\ c & d \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}}_{U} \underbrace{\begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix}}_{\Sigma} \underbrace{\begin{bmatrix} \cos\phi & \sin\phi \\ -\sin\phi & \cos\phi \end{bmatrix}}_{V^T}$$

Now let us recall formula (246), which explains that the transpose of a matrix $A$ has the transposed singular value decomposition. So what can we say about the "inverse" of a matrix? The reason why I put the word "inverse" in quotes is that we are in the rectangular matrix case, where a proper inverse most likely does not exist. However, if it did, then from formula (243) we would expect:

$$\text{“}A^{-1}\text{”}(\boldsymbol{u}_i) = \frac{\boldsymbol{v}_i}{\sigma_i} \tag{259}$$

The goal of the following is to define a notion of "inverse" which satisfies the equation above.

**Definition 22.** *The* ***pseudo-inverse*** *of an $m \times n$ matrix $A = U\Sigma V^T$ is the $n \times m$ matrix:*

$$A^+ = V\Sigma^+ U^T \tag{260}$$

*where* $\Sigma^+ = \begin{bmatrix} \frac{1}{\sigma_1} & 0 & 0 & 0 & \dots \\ 0 & \ddots & 0 & 0 & \dots \\ 0 & 0 & \frac{1}{\sigma_r} & 0 & \dots \\ 0 & 0 & 0 & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}.$

The pseudo-inverse coincides with the inverse in the cases when the latter exists, i.e. $m = n = r$. In general, the pseudo-inverse satisfies the following inverse-like properties:

$$A^+A = V\Sigma^+U^TU\Sigma V^T = V(\Sigma^+\Sigma)V^T = V \left[\begin{array}{c|c} I_r & 0 \\ \hline 0 & 0 \end{array}\right] V^T$$

$$AA^+ = U\Sigma V^TV\Sigma^+U^T = U(\Sigma\Sigma^+)U^T = U \left[\begin{array}{c|c} I_r & 0 \\ \hline 0 & 0 \end{array}\right] U^T$$

If the block matrix in the right-hand sides were just equal to the identity matrix $I$, then $A$ and $A^+$ would be bona fide inverses. However, as it is in the formulas above, the block matrix is simply the projection matrix onto the first $r$ coordinate vectors. And when we conjugate this block matrix by the orthogonal matrices $V$ and $U$, respectively, we conclude that:

$$A^+A = \text{projection onto } C(A^T) \tag{261}$$

$$AA^+ = \text{projection onto } C(A) \tag{262}$$

This happens because the row space (respectively the column space) of the matrix $A$ is spanned by the first $r$ vectors of the matrix $V$ (respectively $U$), as in the Remark in the previous Lecture.

As an example, let's compute the pseudo-inverse of the matrix:

$$A = \begin{bmatrix} 9 & 12 \\ 12 & 16 \end{bmatrix}$$

This matrix has rank 1, so its SVD is computed just like at the end of the previous lecture:

$$A = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \begin{bmatrix} 3 & 4 \end{bmatrix} = \begin{bmatrix} \frac{3}{5} \\ \frac{4}{5} \end{bmatrix} \cdot 25 \cdot \begin{bmatrix} \frac{3}{5} & \frac{4}{5} \end{bmatrix} \quad \Rightarrow \quad A^+ = \begin{bmatrix} \frac{3}{5} \\ \frac{4}{5} \end{bmatrix} \cdot \frac{1}{25} \cdot \begin{bmatrix} \frac{3}{5} & \frac{4}{5} \end{bmatrix} = \frac{1}{625} \begin{bmatrix} 9 & 12 \\ 12 & 16 \end{bmatrix}$$

Pseudo-inverses have a nice application to least squares: remember that the closest approximation to the system $A\boldsymbol{v} = \boldsymbol{b}$ is $\boldsymbol{v} = (A^T A)^{-1} A^T \boldsymbol{b}$, but we always made the caveat that the columns of $A$ need to be independent in order for $A^T A$ to be invertible. If $A$ has dependent columns, while we cannot write $(A^T A)^{-1}$, we can use the pseudo-inverse to give us the following solution:

$$\textbf{the closest } A\boldsymbol{v} \textbf{ is to } \boldsymbol{b} \textbf{ is achieved for } \boxed{\boldsymbol{v}^+ = A^+ \boldsymbol{b}} \tag{263}$$

Such a $\boldsymbol{v}$ is not unique, basically because the fact that $A$ has dependent columns implies that its nullspace is non-trivial (so to any particular solution $\boldsymbol{v}$, you could add any vector in the nullspace of $A$, and you would still get a solution). But to see that the particular solution $\boldsymbol{v}^+$ in (263) does indeed satisfy $A\boldsymbol{v}^+ = \text{proj}_{C(A)}\boldsymbol{b}$, just apply $A$ to both sides of the equation in the box. You get:

$$A\boldsymbol{v}^+ = (AA^+)\boldsymbol{b} = \text{proj}_{C(A)}\boldsymbol{b}$$

as a consequence of (262). So $A\boldsymbol{v}^+$ is precisely the projection of $\boldsymbol{b}$ onto the column space of $A$, as required by the geometry of least squares approximation.

Let's now consider one last application of SVD's in the case of square matrices (i.e. $m = n$):

**Definition 23.** *The **polar decomposition** of an $n \times n$ matrix $A$ is:*

$$A = QS \tag{264}$$

*where $Q$ is orthogonal, and $S$ is positive semidefinite.*

The name stems from the fact that in the $1 \times 1$ case, any number $a$ has a polar form (225): the number $e^{i\theta}$ has absolute value 1 so it behaves like an orthogonal $1 \times 1$ matrix, and the number $r \geq 0$ is a positive semidefinite $1 \times 1$ matrix. In general, you can obtain the polar decomposition of a matrix from its singular value decomposition, as follows:

$$A = U\Sigma V^T = \underbrace{(UV^T)}_{Q} \cdot \underbrace{(V\Sigma V^T)}_{S} \tag{265}$$

Indeed, the product of orthogonal matrices is orthogonal (it is key here that $m = n$, i.e. this all applies to square matrices), which means that $Q := UV^T$ is orthogonal. Moreover, $S := V\Sigma V^T$ is the usual formula for the diagonalization of a symmetric matrix, as we have seen in (232). The fact

that the eigenvalues of $S$, i.e. the diagonal entries of $\Sigma$, are non-negative is precisely equivalent with saying that $S$ is positive semidefinite. The eigenvalues of $S$ are the singular values of $A$.

## **MIDTERM 2** *(April 30)*